

FIDELITYLOOP: SUPPLEMENTARY EVIDENCE AND REPRODUCTION

A REPRODUCTION AND CLAIM-EVIDENCE MAP

The source release includes synthetic interface checks, historical facade compatibility checks, and a standalone E1 accounting verifier. `REPRODUCING.md` gives the commands and expected outputs. In the E1 package, `python -B verify.py` checks the manifest and recomputes 169 resource/cost values from reduced records. Its ten tests run with `python -B -m unittest`; physical-run acceptance is inherited.

The remaining `scripts/`, `docs/`, and `artifacts/` paths in this supplement identify the separately retained full experiment archive. Its `scripts/maxopt_v3_closeout/reverify.py` rebuilds the 36 acceptance reports, and `scripts/maxopt_v3_closeout/analyze.py` generates their tables; both require archived runs outside the compact source release. A recovery addendum restores the root environment snapshot from identical, hash-bound per-run copies and provides a newly generated archive integrity index. The original checksum sidecar remains unavailable; the new index is not presented as the historical original.

- Full-capacity feasibility: nine all2 reports and G-F in `decisions.json`.
- Model accuracy: `prediction_groups.csv`, `comm_on_support.csv`, and `route_confusion.csv`; M1, M1b, and M2 in `decisions.json`.
- H’s negative result: 15 records in `misses.csv`; guard risk-active ticks and IDs in `guard_events.csv`.
- Guard feasibility and cost: nine guard runs in `run_metrics.csv`, with repeat-block comparisons in `paired_costs.csv`.
- Production quality, physical energy, and provider billing: unsupported by this dataset and excluded from the claims.

A.1 Implementation footprint

Table 1 separates policy logic from integration code in the frozen execution sources. We count physical Python lines carrying code, excluding blank lines, comments, and docstrings; shared components are counted once. The audit records file hashes, symbol ranges, and supporting modules,

Table 1. Implementation footprint in code lines. PPO policy code includes 136 lines of reused network implementation. Shared rows are separate from the per-policy counts; backend and training dependencies are itemized in the audit.

Component	Policy	Integration
B-HPA	38	shared
B-PRED	23	shared
B-MMC	29	shared
Deadline guard	142	19
PPO	247	382
Shared baseline support	—	73
Shared runtime control	—	197

including the inherited PPO training code. These counts describe implementation size, not historical porting effort.

`IMPLEMENTATION_FOOTPRINT.json` binds these counts to the frozen sources. The source release provides the counting script and audited source snapshots in `artifact/implementation_footprint/`.

A.2 Per-repeat latency record

Table 2 preserves all three online tail statistics for every formal run, from the same `run_metrics.csv` records used in the main-text summary.

A.3 Evidence provenance and scope

The workload table is generated from `workload_manifest.json`, whose source chain records the 1,404,294-row BurstGPT-derived input, the parent manifest, the forward pool selection, and the fixed thinning threshold. The selected windows have exact prompt overlap counts of 60, 34, and 30 with their old parent windows in steady, recovery, and burst-offline, respectively; this is disclosed as an identity limitation rather than treated as session independence. The source has no session field, and the offline stream is fixed synthetic pressure.

The price table is a revaluation of the same event streams over 27 price cells; it introduces no new rollout or physical run. The guard table counts recorded risk-active ticks, distinguishing target raises from ticks that merely maintain a latched reservation. Two operator-aborted attempts remain in the raw history and are excluded from the 36 formal runs;

Table 2. Per-repeat online p95, p99, and maximum latency in seconds.

Policy	window	repeat	p95	p99	max
all2	steady	1	1.598311	1.613852	1.934303
all2	steady	2	1.600293	1.613611	2.053080
all2	steady	3	1.598100	1.612091	2.338287
all2	recovery	1	1.573470	1.581393	1.587289
all2	recovery	2	1.577489	1.584092	1.587016
all2	recovery	3	1.572175	1.582705	1.583298
all2	burst-offline	1	1.571978	1.582704	1.673667
all2	burst-offline	2	1.573614	1.594019	1.678853
all2	burst-offline	3	1.567986	1.583607	1.675901
all1	steady	1	2.611935	2.632597	3.617594
all1	steady	2	2.606561	2.622125	3.621537
all1	steady	3	2.615337	2.631706	3.626181
all1	recovery	1	1.611564	1.626285	2.564877
all1	recovery	2	1.616560	1.683677	2.581092
all1	recovery	3	1.620432	1.686686	2.577571
all1	burst-offline	1	1.647122	1.704174	1.749571
all1	burst-offline	2	1.612728	1.632177	1.732826
all1	burst-offline	3	1.605751	1.637405	1.729437
H	steady	1	17.001106	19.002347	21.001175
H	steady	2	17.000396	19.001483	21.001591
H	steady	3	17.001118	19.001507	21.002980
H	recovery	1	15.002311	15.008397	15.011078
H	recovery	2	15.002046	15.009679	15.014929
H	recovery	3	15.002045	15.008747	15.012199
H	burst-offline	1	15.003550	15.009867	15.011047
H	burst-offline	2	15.004760	15.046479	15.116333
H	burst-offline	3	15.002258	15.008656	15.009492
H+guard	steady	1	17.001446	19.002137	21.005507
H+guard	steady	2	17.001417	19.001712	21.005507
H+guard	steady	3	17.001367	19.001675	21.004582
H+guard	recovery	1	15.002007	15.008547	15.009813
H+guard	recovery	2	15.002468	15.008545	15.046810
H+guard	recovery	3	15.002523	15.008521	15.031079
H+guard	burst-offline	1	15.001992	15.003827	15.009511
H+guard	burst-offline	2	15.002109	15.005205	15.011248
H+guard	burst-offline	3	15.002959	15.024269	15.046434

the 15 H recovery misses remain in the formal evidence and are not replaced by guarded outcomes.

A.4 Bridge campaign and guard audit

The separate Bridge campaign contains 38 locked cells: an all2 capacity anchor, two hysteresis down-threshold settings with and without the guard, and two frozen PPO policies, each evaluated on recovery and steady windows. It is historical evidence from a different runtime group, not an extension of the E2 pool. The compact policy summary is shown below before the raw-event audit.

The Bridge guard audit joins each recorded risk ID to its request terminal and absolute deadline; it does not rerun the policy or form a counterfactual. Each cell below aggregates three guarded repeats. A risk-active run has at least one nonempty risk-ID set; target raises are averaged over risk-

Table 3. Separate 38-cell Bridge campaign. Entries show recovery/steady P1+ counts, followed by operating cost in the same order.

Policy	P1+ R	P1+ S	occupied R/S	cost R/S
all2 (capacity anchor)	1/1	1/1	4200/4200	4.200/4.200
H-d0	3/3	0/3	1625/1778	1.752/2.135
H-d0+guard	3/3	0/3	1623/1975	1.752/2.314
H-d1	0/3	3/3	1547/1841	1.709/2.208
H-d1+guard	0/3	3/3	1707/1837	1.870/2.226
PPO-C30	1/3	3/3	1387/1991	1.626/2.075
PPO-C	0/3	0/3	1184/1218	2.547/2.935

Table 4. Raw-event guard audit for the 12 guarded cells in the separate Bridge campaign.

Down	window	risk runs	raises/run	misses/360	P1+ pass
d0	recovery	0/3	0	0/360	3/3
d0	steady	3/3	2.0	9/360	0/3
d1	recovery	3/3	2.0	7/360	0/3
d1	steady	0/3	0	0/360	3/3

active runs; deadline misses count late or unfinished offline requests across the three repeats.

B V3 ACCOUNTING AND GATE DETAILS

Completed requests have a completion terminal by the observation cutoff. On-time completion includes equality with the deadline; late completion is strictly after it. Unfinished requests lack a completed terminal at cutoff, and missing or censored requests are not removed from the arrival denominator. G_F requires every all2 online and offline arrival on time. P1 requires 120/120 offline arrivals on time, at least 99% of online arrivals on time, and no unfinished request. P1+ additionally requires every online arrival on time. These definitions follow `scripts/maxopt_v3/acceptance.py`; the extension’s strict qualification uses the same timely/unfinished distinction.

B.1 Replay model specification

The main text lists the frozen O/C/C30/E/V parameters and controlled contrasts. This section gives their construction and provenance. E2 reuses the frozen C/C30 pair.

Service construction. For input length i , output budget o , and local concurrency c , O assigns service duration $c(i/1024 + o/128)$ seconds, using assumed rates of 1024 input and 128 output tokens/s. C replaces this formula with the median measured wall time per homogeneous input/output/concurrency cell. The replay interpolates linearly

in input length between 128, 512, and 2048 tokens, and in concurrency between one and four, with output budgets restricted to 32, 128, or 256 tokens. A dedicated 256-input/256-output, concurrency-four measurement gives 3.098 s. Each running request accumulates progress at the reciprocal of its current-concurrency service estimate; this is a mixed-workload approximation, not token-level vLLM simulation. C30, E, and V share this lookup unchanged.

Lifecycle construction. O’s 30 s startup is the original assumed constant. C’s 69.219 s proxy is the median of 18 launch-to-ready observations collected during two-instance initialization in nine development runs. The shared historical 7.059 s shutdown proxy comes from nine joint-cleanup durations; the 69.363 s setup proxy is the median of nine setup-barrier durations. C30 restores the 30 s assumption while holding C’s other fields fixed; it is not a new fitted startup estimate.

The earlier empirical lifecycle model separated window restarts from setup and normal scale-in from terminal cleanup. Medians within each run, followed by a median across runs, gave startup/shutdown values of 25.196/1.937 s. V searched the nine combinations of startup multipliers {0.9, 1.0, 1.1} and shutdown multipliers {0.8, 1.0, 1.2} around that center, clipped below by the corresponding observed minima. It minimized equal-window L on the earlier development runs; leave-one-window-out and leave-one-repeat-out checks preceded the final fit. The frozen V uses 25.196/2.325 s and is distinct from the later V' refit.

Before the held-out study, E alone was adapted to the A100 deployment: medians from six development restarts gave 33.497/2.224 s, followed by six confirmation restarts. The service lookup, other models, and selected policy were unchanged; no held-out outcomes entered this adaptation. Thus V–E compares the historical refit with the deployment-adapted model, rather than isolating a fitting-method effect. Windows begin with both instances ready; the inherited setup proxy is not fitted through the window loss. The accompanying `REPLAY_MODEL_SPECIFICATION.json` retains the full lookup, unrounded parameters, and source hashes.

B.2 Guard implementation map

The frozen `scripts/maxopt_v3/policy.py` implements the following causal projection and state update. The lookup is a conservative estimate, not a proved upper bound, and infeasible work is still reported rather than guaranteed.

- `validate_observation` (lines 20–47) accepts only released queues, device states, state ages, and running-job dispatch times. Future arrivals, true readiness, and completion times are not policy inputs.

- `duration/remaining/delay` (70–90) uses concurrency four. A running job receives its estimated residual if positive, otherwise a full-service fallback. Active/draining devices have zero added delay; an off device needs the frozen safe startup time. A starting device uses the remaining safe-start interval, then the maximum-start interval (both floored at one second), then restarts the positive safe-start estimate if both are exceeded. A stopping device adds remaining shutdown, or a full shutdown fallback, before startup.
- `projected` (92–126) chooses the k devices with least lifecycle delay, with device-ID ties, and builds four slots each. Offline jobs are sorted by deadline and ID. Without reservation, current online backlog precedes offline work. With reservation and online backlog, offline jobs are serialized from the earliest running offline slot, or the earliest slot if none is running. Without online backlog they use all slots. No future arrivals are projected.
- `decide` (128–158) first obtains H’s target b and intersects latched risk IDs with pending offline IDs. It adds IDs with projected finish $F_j(\max(1, b), r) + m + 1 \geq d_j$, where $m = 5$ s. With active risk and scaling enabled, it tests $k \in \{1, 2\}$ using $F_j(k, r) + m \leq d_j$ for every pending offline job. The target is $\max(b, \min k)$ when feasible; otherwise it is two and `infeasible` is set. A raise bypasses cooldown and updates H’s target and last-change time. Safe re-estimation alone does not clear a latched ID; dispatch or removal from the pending queue does. Reservation is enabled only while risk is latched and the reservation option is enabled.
- `dispatch_kind` (160–172) never preempts running work. With both queues nonempty and a reservation active, an already-running offline job leaves free slots online-first. Otherwise online takes the next slot only when its head request’s estimated slack is at most m and no larger than the offline head’s slack. With no online queue, offline dispatch is unrestricted by the single-slot reservation; without an active reservation, online has priority.

Table 5. Accepted V3 cost ledger. Costs are means over three repeats; the final two columns are savings percentages.

Window	all1	H+guard	window	deployment
steady	2.1029	1.6411	21.96	21.41
recovery	2.1029	1.0895	48.19	37.82
burst-offline	2.1029	0.8815	58.08	57.34

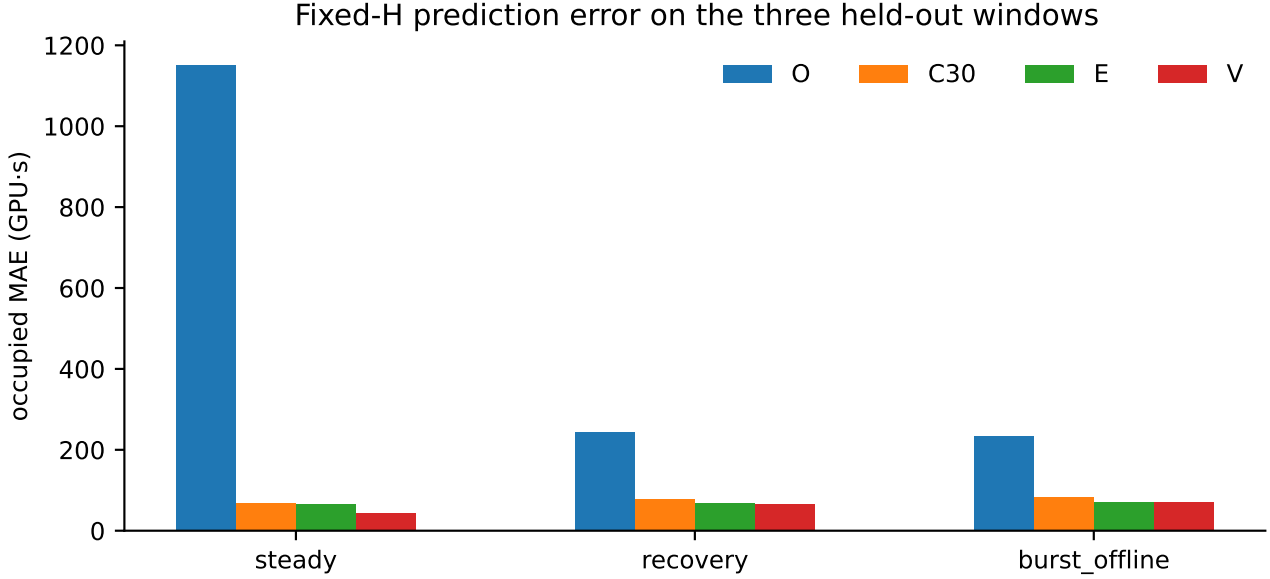


Figure 1. Occupied prediction error on fixed H. C30, E, and V have lower error than O on all three windows; the main-text table also includes C.

Table 6. Gate interpretation for the locked V3 route; M3 is diagnostic.

Gate	observed result	interpretation
G-F	all2: 9/9	full-capacity feasibility
M1	E: 87.50% reduction	occupied-resource gate
M1b	V: 34.24% <i>L</i> reduction	environment caveat
M2	O fail; C/C30/E/V pass	matched-service fidelity
M3	E: 26.77%	diagnostic, no veto
P1+	H+guard: 9/9	online/offline feasibility
P2	mean: 42.74%	window-cost gate

B.3 API-delay sensitivity in CPU replay

We replay the four frozen rule policies (all1, all2, H, and H+guard) on the three original windows under model E, varying only the fixed API service delay over {1, 2, 5, 10, 20, 40, 60} s. API capacity remains eight concurrent requests, with the same 10 s routing wait, prices, and policy parameters. All 12 five-second baselines reproduce the frozen predictions, yielding 84 deterministic CPU scenarios in total.

At the tested delays of 1, 2, 5, and 10 s, both H+guard and all1 satisfy P1+ in all three windows. Their equal-window cost savings are 45.07%, 45.07%, 45.08%, and 43.76%, respectively. At 20 s, H+guard misses online deadlines in recovery and steady; at 40 and 60 s, online misses occur in all three guarded windows. Longer API service occupies the finite API slots and changes subsequent queueing, routing, and GPU control. These results test sensitivity to the

timer assumption; they are CPU predictions, not additional physical runs or an empirical calibration of a commercial API.

C V4 IDENTITY AND EVIDENCE LEDGER

The V4 source records supplement, rather than replace, the 36 frozen V3 runs. Table 7 distinguishes new execution, historical reuse, and gate roles. All 93 accepted identities are included once; the full-strength W4’s 18 gate-stopped policy cells are not counted as executions. W4-L’s p075 and p050 gate batches are qualification evidence, not additional policy repeats.

Strict qualification requires at least 99% online on-time completion, all offline jobs completed by their deadlines, and every expected request completed without censoring or missing work. Offline counts use each input’s actual expected denominator; formal comparison windows here contain 120 jobs, whereas development windows may contain a different count or none.

The generated V4 rows are bound to the CPU closeout’s `paper_data/TABLE_MANIFEST.json`, `run_metrics.csv`, `window_summary.csv`, `equal_window_summary.csv`, and `w4l_pairs.csv`. The source root is `artifacts/max_optimization_v4_20260923/cpu_closeout_20260927/`. Repeated requests in these CSVs are evidence about a run, not independent replications. V3’s P2 uses the equal average of within-window ratios; the W4-L headline instead uses a ratio of

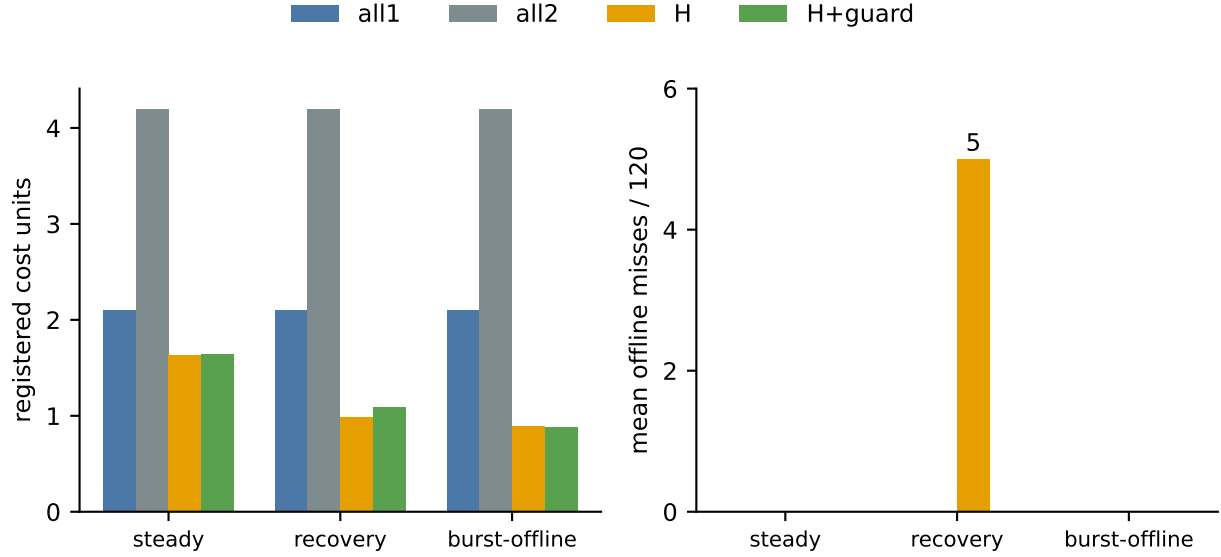


Figure 2. Window cost and offline deadline misses, averaged over three repeats per arm/window. H averages five recovery misses (15 total); all other bars are zero. The guarded arm has no misses while reducing scenario cost relative to all1.

Table 7. V4 coverage and strict qualification. W4-L p050 fresh contains all four policies; its 15/24 total must not replace their per-policy counts.

Workstream	fresh	reused	identities	strict	role
W1 baselines	27	0	27	27/27	comparison
W2 Llama	18	6	24	21/24	comparison
W4 full load	6	0	6	0/6	gate
W4-L p075 gate	6	0	6	0/6	gate
W4-L p050 gate	6	0	6	6/6	gate
W4-L p050 fresh	24	0	24	15/24	comparison
Total	87	6	93	—	mixed roles

equal-window means. The two statistics are reported under their own definitions and are not numerically pooled.

C.1 Baseline observation and selection contract

B-HPA updates every 60 seconds using the preceding queue observations; its registered grid is $q_{\text{target}} \in \{4, 8, 16\}$ and stabilization $\in \{120, 300, 600\}$ seconds. B-PRED uses Holt level/trend updates with $\alpha \in \{0.2, 0.4, 0.6\}$, $\beta \in \{0.1, 0.3, 0.5\}$, horizon $\in \{60, 120, 300\}$ seconds, and margin $\in \{0, 1, 2\}$ (81 candidates). B-MMC’s utilization ceiling is drawn from $\{0.70, 0.80, 0.85, 0.90\}$. Its service lookup is shared with E to isolate the target rule; this does not provide future information or make it an oracle. Development ranking places feasibility before equal-window cost; the preregistered center is retained if a family has no qualifying candidate. None of the three formal arms includes the guard.

All 9/81/4 candidates, respectively, failed development qualification. The predeclared fallback therefore fixes B-HPA at queue target 8 and stabilization 300 seconds, B-PRED at $\alpha = 0.4$, $\beta = 0.3$, horizon 120 seconds and margin 1, and B-MMC at $\rho_{\text{max}} = 0.80$. The original development feasibility field incorrectly rejected windows containing zero offline jobs. Recalculation with each window’s actual offline denominator still leaves all 94 candidates ineligible: each independently violates the 99% online criterion. Thus the fallback is unchanged, but the original field is not treated as authoritative. The subsequent fixed-center validation repairs the preceding-60-second interval to $[t - 60, t)$ and checks the same three centers. It performs no grid search and reads no formal or locked-test outcome; it is semantic validation, not retuning.

The runtime continuation is a comparability limitation, not an exclusion rule. Historical W1 cells 001/002 and continuation r14 remain in the nine-per-arm denominators. Baseline-

Table 8. W1 online end-to-end p99 latency (seconds): minimum–maximum of the three completed-request per-run quantiles. These are repeat ranges, not confidence intervals or pooled request quantiles.

Policy	steady	recovery	burst
B-HPA	19.00–19.00	15.00–15.00	1.66–1.81
B-PRED	1.61–1.68	1.58–1.59	1.60–2.70
B-MMC	19.00–19.00	15.00–15.00	1.58–2.01

Table 9. Sequential all2 qualification. Front/back list timely offline jobs out of 120 for each of three repeats. Gates are excluded from the fresh policy comparison.

Intensity	front	back	strict	action
1.00	35/34/35	37/37/37	0/6	stop
0.75	113/113/113	79/79/79	0/6	lower load
0.50	120/120/120	120/120/120	6/6	lock; fresh 24

versus-V3 comparisons are descriptive; their repeat indices do not establish randomized pairing, and request-level re-sampling would not correct the confound.

C.2 Failure records that constrain the claims

W2 H recovery cells 042/043 each contain 116 completed jobs out of the original 120; four unfinished jobs per cell remain in the denominator. Guard recovery cell 049/r2 completes 120 but has only 119 timely jobs. These facts explain why 24/24 technical validity, 23/24 historical scientific acceptance, and 21/24 strict qualification are not contradictory. W2-P is not established by a technical or historical aggregate pass.

For W4, the full-strength all2 gate stops the policy matrix after six scientific failures. The p075 amendment gate also fails, despite eventual completion of all offline work. The first feasible p050 gate permits, but does not substitute for, 24 fresh executions. In those fresh records H front/r1 and front/r3 each leave two offline jobs unfinished; H back/r2 leaves four. all1’s six failures remain part of the official negative aggregate. No scientific failure is relabeled as an operator abort, deleted, or replaced by a guarded result.

C.3 Source and intensity provenance

The Azure input is the conversation CSV at immutable repository commit 207bed67dd10090b28ad4f745b2cfd41a11aace4, with SHA-256 2f1e5b666d4e3055fdbba98598ce2ec307767b9064e03e2fa46676dbcc7d0bf8. Its three columns are timestamp, context tokens, and generated tokens; only timestamps construct the replay arrivals. The code trace is archived but not substituted.

Table 10. W4-L 0.50 fresh comparison. Equal-window means preserve all three repeats in each window. Only all2 and H+guard qualify strictly in all six runs. Costs are scenario USD, not measured provider bills.

Policy	strict	occupied	window	deployment
all2	6/6	4200.0	4.2000	4.3204
all1	0/6	2101.9	3.5248	3.6093
H	3/6	2958.9	3.0428	3.1186
H+guard	6/6	2992.4	3.0675	3.1366

Table 11. W4-L p050 fresh equal-window segment means. Costs are scenario USD; occupied time is GPU seconds. The final 600 seconds supply about 81.96% of guard’s absolute whole-window saving against all2.

Policy	seconds	occupied	cost	API cost
all2	0–1500	3000.0	3.0000	0.0000
all2	1500–1800	600.0	0.6000	0.0000
all2	1800–2100	600.0	0.6000	0.0000
all1	0–1500	1501.9	2.8609	0.5779
all1	1500–1800	300.0	0.3639	0.0039
all1	1800–2100	300.0	0.3000	0.0000
H	0–1500	2738.8	2.8034	0.0511
H	1500–1800	181.2	0.1864	0.0000
H	1800–2100	38.8	0.0530	0.0000
H+guard	0–1500	2728.1	2.7957	0.0536
H+guard	1500–1800	202.4	0.2081	0.0000
H+guard	1800–2100	61.9	0.0638	0.0000

The W4-L integer-SHA rule preserves the parent source row and timestamp of every retained online arrival and keeps the offline stream unchanged. Thus p050 means a locked online inclusion intensity, not half the offline work or a rescaled time axis. Neither p050 nor the unused optional p025 is a new independent sample of Azure production workloads.

Independent CPU checks regenerate inputs and request metrics from raw records, and bind worker events to request identity and lifecycle generations. The supplemental provenance package closes the three missing original-source links; its isolated replay test suite passes 107 tests. This is verification of the available artifact, not new GPU evidence. The original V3 convenience-snapshot warning remains historical and is not silently rewritten.

C.4 Time-segment cost accounting

The first interval is the nominal 1,500-second source-arrival phase, not the whole replay. The next interval still contains 20 offline arrivals per window, and the back window’s closed endpoint retains an online arrival at 1500 seconds. The [1800, 2100) interval extends observation through the remaining deadline horizon. These intervals are accounting

Table 12. W2 fixed-H model error; O_x and E_x have the same values at the shown precision. Each window first averages its two physical repeats.

Window	L (both models)	occupied MAE
steady	0.01170749	73.09045
recovery	0.00059876	1.44829
burst-offline	0.00032530	0.89862
equal-window	0.00421052	25.14579

partitions, not claims that every non-source second is empty or that the entire last 600 seconds represent a production steady state.

D CPU MODEL EVIDENCE

D.1 W2: within-model gates and common support

The accepted W2 analysis aligns 48 O_x/E_x predictions with 24 physical sources. It preserves the six historical all2 identities exactly once. Both models are derived from development calibration only: O_x uses a cell median and E_x a cell maximum. The simulator consumes the two-GPU aggregate lookup, not the per-GPU metadata. Both have startup/setup 29.213463 seconds and shutdown 1.917728 seconds. Actual guarded execution uses its fixed E_x plan; each counterfactual predictor uses its corresponding model with the same policy and safety settings. No formal outcomes fit these service or lifecycle values.

For horizon T , two devices, and N total arrivals, the registered loss is

$$L = \frac{0.5|\Delta G_{\text{occupied}}| + 0.25|\Delta G_{\text{active}}|}{2T} + \frac{0.25|\Delta Q|}{TN}, \quad (1)$$

where Δ is prediction minus observation, $T = 2100$ s for formal windows, and N includes all online and offline arrivals. Q is the integrated released-but-undispatched queue in request seconds. Unfinished jobs remain in N and contribute queue time while queued up to cutoff. Losses are averaged over repeats within each window and then equally across windows, not over pooled requests or matched-local support. The W3 development horizon is 900 s. This prediction-error objective is distinct from monetary scenario cost. The W2 main gate is L reduction $\geq 20\%$ on H together with per-window occupied MAE $\leq O_x + 21$ GPU seconds. The inherited occupied-improvement threshold is a separate diagnostic, not an alternative route to a pass.

M2 uses the intersection of actual-GPU, O_x , and E_x completed-local IDs. Pairwise support is retained separately even though it coincides with common support in these records. Fixed and dynamic M2 criteria pass, with minimum common support 206 and maximum per-run common MAE 0.02963983 seconds. This is not a whole-stream fidelity

claim: equal-window coverage over all arrivals is 100% for all1/all2, 51.2417% for H, and 58.8975% for guard. In H’s two steady repeats, only 454/457 requests are common-local, respectively, covering 29.0281%/29.2199% of arrivals and 53.4747%/54.0189% of actual-local completions. Targets disagree with the actual controller at 1401/2100 and 1397/2100 ticks despite matching switch counts of 28. Equal-window target disagreement is 22.2063% for H and 0.6190% for guard.

The zero main-metric improvement is not accidental model duplication: 14,324 requests have different O_x/E_x predicted completion times. Those small service changes do not sufficiently change the registered resource and queue metrics. The r2 gate-only repair restores the pre-registered safety conjunction; the 48 predictions and numeric tables remain byte-identical to r1. Independent recalculation and boundary tests accept this implementation while retaining the scientific negative result. Evidence resides in `w2/decisions.json`, `w2/prediction_errors.csv`, `w2/route_confusion.csv`, and `w2/target_comparison.csv` under the CPU close-out root stated above.

D.2 W3: bounded refit and an unmet improvement gate

The W3 development bundle contains a preformal drift record and two historical Qwen engineering canaries (sparse and recovery), each with one observation. Their model bytes, engine configuration, and platform match the frozen formal deployment. The canaries contain 41 and 1440 ordinary completed GPU service pairs, respectively; health probes are excluded. Their request IDs are disjoint from formal IDs. These dynamic records are not independent fixed-concurrency experiments or additional formal repeats.

A nine-cell grid varies startup by 0.9/1.0/1.1 and shutdown by 0.8/1.0/1.2 around E, subject to preformal observed floors of 24.958327 and 1.897256 seconds. The development loss uses the same dimensionless definition above at its 900-second horizon, then averages the two windows equally. The selected V' has startup 30.146928 seconds and shutdown 2.224161 seconds; only startup differs numerically from E. Setup, the service lookup, policy, guard model E, and all guard safety settings remain frozen. Prepare, fit, and evaluate locks are ordered and hash-bound; formal target/event/occupied values do not enter selection. Nevertheless, the prior V3 aggregate outcomes were already known, so the procedure cannot be described as a new blind study.

All 36 accepted formal identities enter evaluation. Twelve unique deterministic replays per model are compared with the physical repeats, producing 72 model-by-run error records; these are not 72 new executions. E’s summary,

Table 13. W3 fixed-H replay. Each window averages three physical repeats; occupied MAE is in GPU seconds. Steady L worsens. All occupied safety lines pass, but the equal-window improvement of 5.02% misses the 10% gate.

Window	$E\ L$	$V'\ L$	E occupied	V' occupied
steady	0.012390	0.014516	66.34	70.67
recovery	0.012889	0.010865	66.95	66.95
burst-offline	0.015303	0.013164	70.55	70.55

requests, and events exactly reproduce all 12 frozen V3 prediction groups. The primary fixed-H equal-window loss is 0.0135273755 for E and 0.0128485693 for V' , a reduction of 5.018019%. The conjunctive improvement-and-safety gate therefore fails despite three passing $E + 21$ safety checks. Guard’s 4.073089% reduction is descriptive and cannot replace the primary H result; all1/all2 are unchanged.

A technical correction in evaluation maps accepted short window names to their full locked identifiers. It precedes prediction output and outcome comparison, leaves the fitting grid/loss/tie-break and frozen model unchanged, and is retained in `w3/evaluation_technical_amendment.json`. Independent review reproduces the raw ledgers, selection, loss, and representative complete simulator objects. The accepted conclusion remains negative with two-canny, retrospective-knowledge, and lifecycle-only scope limits. Evidence includes `w3/prepare/analysis_plan.json`, `w3/fit/frozen_model.json`, and `w3/evaluate/result.json`; no new GPU run is introduced.

E E1: SAME-RUNTIME SUPPLEMENTARY COMPARISON

E1 compares frozen H+guard and B-HPA on the previously observed `locked_test_steady_v3` and `locked_test_recovery_v3` inputs without revising the V3/V4 locks. All 110 formal lifecycle generations share the two-A100 Qwen2.5-7B-Instruct FP16 runtime, model closure, service lookup, health probe, synthetic API, and accounting contract. Both instances start physically ready with empty queues; adapter targets remain B-HPA=0 and H+guard=2. These targets and the intrinsic dispatch rules belong to the complete policy treatments, so E1 does not isolate the causal contribution of the guard module.

E.1 Identities, feasibility, and retained smoke outcomes

The formal matrix contains one all2 capacity run per window and three fresh repeats of each comparison policy per window: 2 capacity plus 12 comparison identities. Each steady run contains 1,444 online and 120 offline arrivals;

each recovery run contains 334 online and 120 offline arrivals. Independent raw recomputation finds all 14,126 formal arrivals completed on time, with zero late, failed, or unfinished requests. Both capacity runs pass G_F and every comparison run passes P1/P1+. Table 14 retains all 14 identities; the capacity runs are qualification evidence, not extra B-HPA or guard repeats.

The 17 entries include three 90-second nonformal smokes. B-HPA a1 is `TECHNICAL_INVALID`: its lifecycle callback rejected ready after forced stopping, leaving four offline arrivals unfinished. After repair, a2 records 68/68 online completions on time but 0/4 offline completions at cutoff; its `NEGATIVE_RESULT` and false gate remain intact. The wrapper’s `PASS` denotes technical integration only. Guard’s smoke completes all 68 online and four offline arrivals on time. None enters the formal comparison. The raw `formal_accepted=false` field is not overwritten; scientific qualification is independently derived from request and lifecycle records.

E.2 Resource use and cost boundary

For window w , let $\overline{C}_{g,w}$ and $\overline{C}_{h,w}$ be the arithmetic means across the three H+guard and B-HPA repeats. E1 reports $R_w = 1 - \overline{C}_{g,w}/\overline{C}_{h,w}$ only after both arms qualify in all three runs, then $R = (R_{\text{steady}} + R_{\text{recovery}})/2$. This is the equal average of within-window ratios, not a pooled ratio or the average of six per-pair percentages. The fixed repeat blocks support descriptive paired differences, not confidence intervals or population-level significance. Window scenario-cost reductions are 32.5975% and 49.3598%, hence 40.9786% equally weighted. Full-deployment reductions are 31.7650% and 47.8629%.

Occupied time spans `launch_issued` through `verified_release`, including starting and stopping, clipped to each phase. Window occupied-time reductions are 50.8839% and 56.1240%; the corresponding mean savings are 1,224.82 and 1,147.44 GPU seconds. This is local resource occupancy, not total compute across the local and external services. Active time is not GPU compute utilization. Guard’s zero cleanup cost means that release already completed within the window, not that release or setup is free.

Repricing the fixed ledgers makes the cost boundary explicit. Let ΔC_G , ΔC_A , and ΔC_E denote mean H+guard minus B-HPA GPU, API, and lifecycle-event costs in scenario USD. Their steady values are $(-1.2248, 0.3862, 0.0340)$; recovery gives $(-1.1474, 0.0930, 0.0320)$. With GPU price multiplier x , a common API input/output-budget multiplier y , and fixed lifecycle-event prices, the cost difference is $\Delta C(x, y) = x\Delta C_G + y\Delta C_A + \Delta C_E$. Neither arm incurs missed-job penalties. Since $\Delta C_A > 0$, H+guard costs less

Table 14. All E1 formal identities. Occupied time is window GPU seconds; costs are scenario USD. API is the percentage of all online arrivals, p99 is each run’s completed-online end-to-end quantile in seconds, and S/D counts window starts/shutdowns. Every row completes all online and offline arrivals on time.

Window	Policy	Repeat	Occupied	Window cost	Full cost	API (%)	p99	S/D
steady	all2	gate	4200.0	4.2000	4.2700	0.00	1.616	0/0
steady	B-HPA	1	2407.0	2.4678	2.5389	6.37	19.002	2/3
steady	B-HPA	2	2407.1	2.4696	2.5398	6.58	19.005	2/3
steady	B-HPA	3	2407.2	2.4680	2.5383	6.37	19.003	2/3
steady	H+guard	1	1266.0	1.7611	1.8339	53.25	19.002	13/15
steady	H+guard	2	1140.5	1.6153	1.6842	50.83	19.002	13/15
steady	H+guard	3	1140.3	1.6151	1.6794	50.83	19.002	13/15
recovery	all2	gate	4200.0	4.2000	4.2758	0.00	1.586	0/0
recovery	B-HPA	1	2044.6	2.0715	2.1415	11.68	15.002	1/2
recovery	B-HPA	2	2044.4	2.0713	2.1386	11.68	15.002	1/2
recovery	B-HPA	3	2044.4	2.0713	2.1439	11.68	15.002	1/2
recovery	H+guard	1	1102.2	1.2576	1.3279	60.18	15.008	12/14
recovery	H+guard	2	794.5	0.9447	1.0110	58.98	15.008	11/13
recovery	H+guard	3	794.4	0.9446	1.0104	58.98	15.008	11/13

Table 15. E1 mean phase costs (scenario USD), three repeats per comparison arm/window. Full includes setup, window, and cleanup.

Window	Policy	Setup	Window	Cleanup	Full
steady	B-HPA	0.0675	2.4685	0.0031	2.5390
steady	H+guard	0.0687	1.6638	0.0000	1.7325
recovery	B-HPA	0.0670	2.0714	0.0030	2.1413
recovery	H+guard	0.0675	1.0489	0.0000	1.1164

when

$$y < y^*(x) = -\frac{x\Delta C_G + \Delta C_E}{\Delta C_A}.$$

At $x = 1$, the API break-even multipliers are 3.08 and 11.99 (3.09 and 12.02 for full deployment). At $y = 1$, window savings require GPU price multipliers above 0.343 and 0.109. Lower local GPU occupancy therefore need not imply lower cost: the boundary depends on external-service prices. This analysis holds routing and lifecycle events fixed, without retuning either policy.

E.3 Routing, lifecycle, and queue trade-offs

The online API fraction uses all online arrivals as denominator, not only local completions. Its mean rises from 6.44% to 51.64% in steady and from 11.68% to 59.38% in recovery. Table 16 retains the additional lifecycle events and queue exposure. Queue exposure sums each request’s actual release-to-dispatch time, or release-to-cutoff time if undispached, in request seconds. It is neither a queue-latency p99 nor a new value of the historical prediction loss L . Tail latency uses completed online requests; the all-arrival feasibility checks above independently rule out hidden formal

Table 16. E1 trade-offs. S and D are mean window starts and shutdowns; Q is mean queue request-seconds. p99 ranges span three per-run completed-online quantiles (seconds), not confidence intervals or pooled-request quantiles.

Window	Policy	S	D	Q	p99 range
steady	B-HPA	2.00	3.00	1674.9	19.002–19.005
steady	H+guard	13.00	15.00	12884.7	19.002–19.002
recovery	B-HPA	1.00	2.00	751.1	15.002–15.002
recovery	H+guard	11.33	13.33	9429.1	15.008–15.008

failures.

The two known inputs and three repeats per cell establish a bounded policy comparison. Greater API use and queue exposure rule out all-dimensional dominance; quality-equivalent API substitution, energy, provider bills, and new-trace generalization remain unmeasured.

E.4 Independent recomputation and artifact links

The accepted table source is docs/maxopt_v4_r3_p5_20260929/ANALYSIS/P5_RAW_v2/: FORMAL_RUNS.csv retains 14 formal identities, ATTEMPTS.csv all attempts, PAIRS.csv six repeat blocks, and AGGREGATES.json the qualified estimands. COMPARABILITY.json binds plans, runtime, and permitted policy differences. In the parent documentation directory, REVIEW_B.json and REVIEW_B_V2.json bind accepted output hashes; P6_GENERATE_E1_TABLES.py and P6_TABLE_MANIFEST.json reproduce and bind the displayed rows. The standalone E1 package reproduces the resource and price-boundary calculations from portable reduced records; its verify.py checks the manifest and

full-precision reference results.

F OFFLINE DECISION-CALL MICROBENCHMARK

This microbenchmark reuses actual states from all 16 technically valid same-runtime comparison runs, including the B-HPA smoke with four unfinished offline jobs; the original technically invalid a1 is excluded with its failure retained. Independent recovery matches the full input and hidden pre/post-state of all 29,580 policy ticks, including guard latched IDs and B-HPA's complete queue history. Remaining work is the frozen policy's surrogate, not GPU progress inferred from future completions.

Before timing, a fixed rule chooses the earliest available tick per run and queue stratum: empty, intermediate, or maximum observed queue size; B-HPA also separates update and hold ticks. Empty strata stay empty. The resulting 60 snapshots are retrospective cases, not random production or held-out samples. Each receives 100 warmup and 1,000 timed calls of the original sealed `adapter.decide`, including its existing validation. Every call uses a newly restored adapter and is checked against the original decision; restoration, copying, and checks are outside `perf_counter_ns` timing. All 60,000 timings, including slow values, are retained; they are repeated measurements of 60 states, not 60,000 independent physical experiments.

The host is an Intel Xeon E5-2682 v4 (2.50 GHz), eight logical CPUs, with CPython 3.13.5, distinct from the GPU execution environment. CPU use is not exclusive; affinity is unchanged and frequency policy is unavailable. Table 17 reports all groups. Formal guard's largest single-snapshot p99 is 120.105 μ s in steady and 92.39661 μ s in recovery; its largest individual observed times are 157.491 and 841.877 μ s. The latter is retained rather than hidden by a quantile. These are neither pooled p99s nor production-run tail estimates.

The interval excludes `dispatch_kind`, admission, the full controller loop, replay/restoration, logging, I/O, worker and GPU execution. It measures offline decision-call cost on this CPU, not complete-controller or online end-to-end overhead, and cannot establish absence of online tail interference. The recovery and timing sources are `ANALYSIS/E2_RECOVERY_v1/` and `ANALYSIS/E2_BENCHMARK_v1/` under `docs/maxopt_v4_r3_p5_20260929/`; the latter preserves all raw samples. `REVIEW_E2.json` binds the independent audit, and `P6_E2_TABLE_MANIFEST.json` binds this presentation.

G E2 POLICY EVIDENCE

The E2 handoff is independently verified by `tools/verify_portable_32_cpu.py`. Its package SHA256 is 079e2b7e146b1706c2500ac9a3c4a5ff0c4310885cc013907e9b86c754399aa3; the extracted package contains 32 cells, 36 attempts, 20 strict P1+ results, 12 scientific negative results, and four retained technical-invalid attempts. No checkpoint was retrained or selected after physical execution. The all2 reference is one cell per window; each PPO cell is a repeated execution of the same frozen checkpoint, not a new training seed.

The physical package stores both C and C30 predictions for each cell. Comparing the frozen prediction labels with the final physical labels gives 32/32 feasibility-direction agreement for C and 20/32 for C30. Window-cost MAE is 0.0804 and 0.4657 scenario USD, respectively; these metrics use the same registered observation-window ledger and do not estimate provider billing. The C30 prediction marks C30-3 feasible in both windows, while all six physical repeats fail the offline deadline. This is a false-positive deployment case, not a reason to remove the negative cells.

The package also retains the exact attempt history, source and tensor hashes, request terminals, controller events, life-cycle manifests, budget receipts, and CPU recomputation for every cell. The four technical-invalid attempts remain in the audit trail and are excluded from scientific repeat counts. The evidence is bounded to two locked windows, two A100 GPUs, one Qwen model, five checkpoints, and the declared synthetic-API scenario ledger.

G.1 Frozen PPO training and selection contract

The capacity policy uses a 30-dimensional observation, a shared MLP with two 256-unit ReLU layers, three capacity logits, and a scalar value head. The four-logit offline head is masked to its sole legal action. Training runs in CPU replay with guard disabled, using C or C30 and 2,100-step episodes and rollouts. Adam uses learning rate 3×10^{-4} and numerical epsilon 10^{-5} ; PPO uses discount 0.9995, GAE parameter 0.999, clip 0.2, four update epochs, minibatches of 64, value coefficient 0.5, entropy coefficient 0.01, and gradient norm limit 0.5. Advantages are normalized and value loss is clipped; the learning rate is constant. These settings are recovered from the frozen configuration, not retuned for the deployment results.

Training uses three development windows with prefix `train_` and suffix `_v2` for the burst-offline, recovery, and steady regimes. Selection uses the corresponding three `validation*_v2` windows with deterministic C30 replay, guard disabled, and P1+ required separately in every window. Eligible checkpoints are ordered by equal-window mean cost, checkpoint index, and seed. E2 reuses

Table 17. Offline `adapter.decide` timings, in microseconds. Median range spans the per-snapshot medians; max p99 is the largest per-snapshot p99, and max sample is the largest individual timing. N counts snapshots.

Kind	Window	Policy	N	Median range	Max p99	Max sample
smoke	steady	B-HPA	3	8.3695–13.0020	24.15627	36.124
smoke	steady	H+guard	3	39.7485–45.1900	61.39790	536.644
capacity	steady	all2	3	2.4390–2.4660	2.97401	19.672
capacity	recovery	all2	3	2.3900–2.4200	2.97409	19.565
comparison	steady	H+guard	9	43.2845–97.4360	120.10500	157.491
comparison	steady	B-HPA	15	12.1775–30.1865	45.79169	389.906
comparison	recovery	B-HPA	15	10.5785–25.8275	38.47390	49.562
comparison	recovery	H+guard	9	43.3315–75.1155	92.39661	841.877

Table 18. Complete E2 policy summary. Costs are means over three PPO repeats or the single all2 anchor in each window. A negative arm can have lower cost while still violating the strict deadline contract.

Arm	training environment	R P1+	S P1+	cost R	cost S	offline outcome
all2	anchor	1/1	1/1	4.200	4.200	120/120
C-2	C	0/3	0/3	2.730	2.950	4–13 on time
C-1	C	3/3	3/3	2.357	2.449	120/120
C30-1	C30	3/3	3/3	2.073	2.084	120/120
C30-2	C30	3/3	3/3	2.051	2.103	120/120
C30-3	C30	0/3	0/3	1.899	2.090	106–114/120

Table 19. Paper labels and frozen checkpoint identities. Steps are those reached by the selected checkpoint. C30-2 is the registered reference.

Label	Archive ID	Seed	Episode	Steps
C-1	C-s101	101	40	84,000
C-2	C-r2	103	100	210,000
C30-1	C30-r2	101	25	52,500
C30-2	C30-ref	101	40	84,000
C30-3	C30-s103	103	75	157,500

the two previously locked Bridge recovery and steady inputs; these are distinct files from the selection inputs. The training contract specifies paired seeds 101–105; Table 19 gives the actual selected checkpoints. The accompanying `PPO_TRAINING_CONTRACT.json` records the configuration, selection-input hashes, and checkpoint identities. Table 19 maps the paper labels to the archived identifiers. In those identifiers, `r2` denotes the second eligible checkpoint, `s101` the seed-101/episode-40 matched pair, and `s103` the best eligible checkpoint within seed 103; the reference is deployed as `ppo_C30_raw`.

G.2 Retrospective E2 lifecycle diagnosis

We diagnose all 32 accepted physical runs and their 24 frozen C/C30 predictions without retraining or selecting new checkpoints. The two prediction configurations differ only in startup time (69.219 s versus 30 s). Window launch-to-ready time exceeds 60 s in 196 of 218 observed

generations; the median of the 30 PPO runs’ startup medians is 87.92 s. These are descriptive lifecycle measurements, not independent experimental replicates.

C-2 exposes the interaction with the 60 s cooldown, which starts at a target change rather than at readiness. In the first recovery run, its first window restart launches at 120.01 s, becomes ready at 223.89 s, and enters draining at 224.01 s without admitting work. The matched C prediction becomes ready at 189.22 s and drains at 190 s, also without admissions. C30 becomes ready at 150 s; although the policy proposes zero capacity, cooldown holds one GPU until 180 s, admitting eight offline requests. Across all six physical C-2 runs, 72 of 73 restarts longer than 60 s enter draining within two seconds of readiness. All 687 offline misses are already overdue before dispatch or remain undischarged at cutoff.

The other negative checkpoint has a milder queueing failure: 60 of C30-3’s 65 offline misses are overdue before dispatch, while all requests eventually complete. Long startup alone does not imply failure: C-1 passes despite 78 of its 84 window restarts exceeding 60 s, retaining capacity to clear its queues. C30-1 and C30-2 also pass, generally keeping a worker available after restarting. These archived trajectories explain a concrete failure pathway and why C detects its feasibility consequence. They do not establish a general ranking of startup models or attribute every physical timing difference to startup alone.

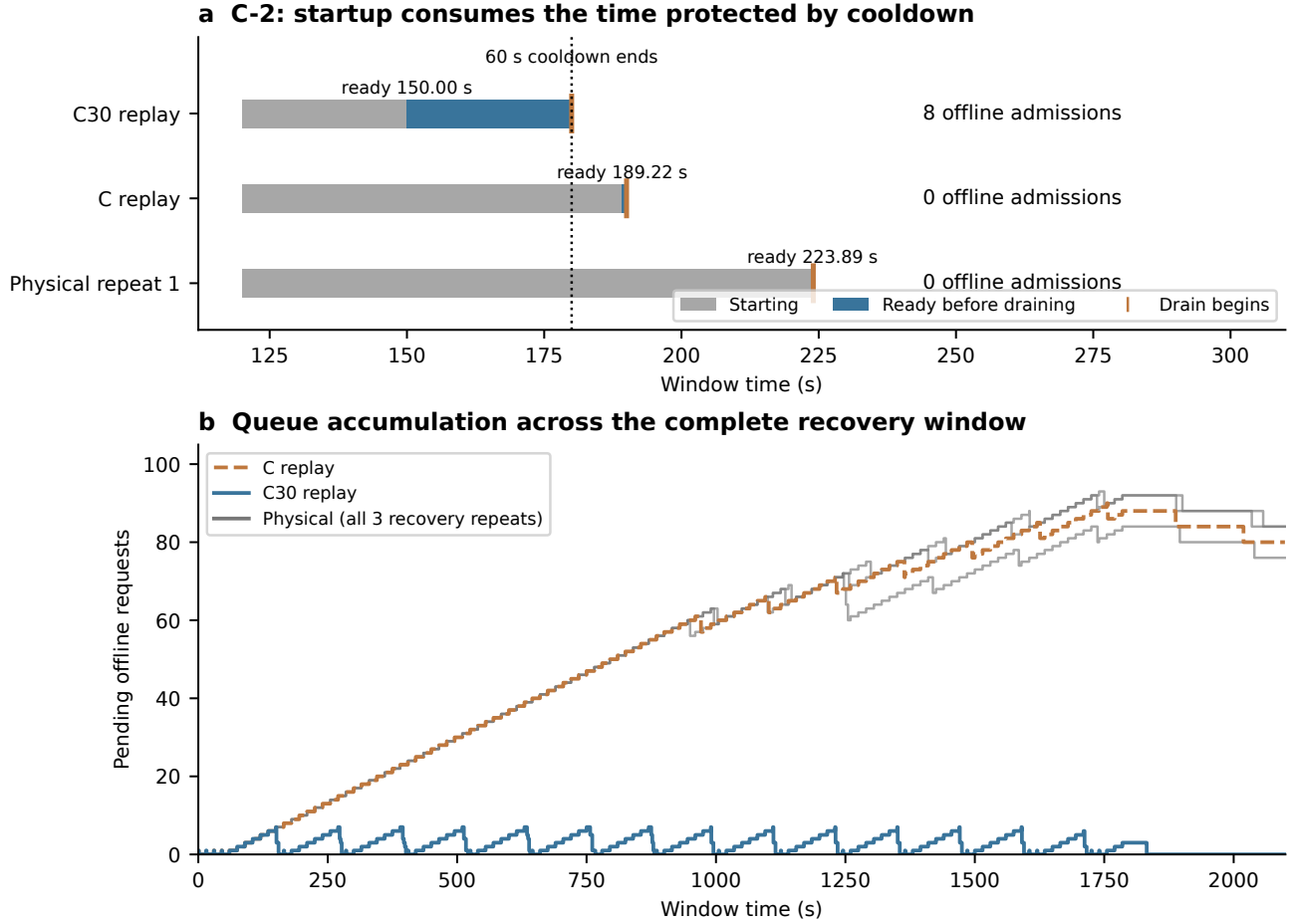


Figure 3. Retrospective C-2 recovery diagnosis. Top: the first window restart of the first formal negative run and its frozen C/C30 predictions; admissions count offline requests dispatched during that instance generation. The dotted line marks cooldown expiry, measured from the target change. Bottom: pending offline queues in both predictions and all three physical recovery repeats. The text summarizes both windows and the passing-policy controls. No new physical runs are introduced.

H E3: SCREENING ADDITIONAL CHECKPOINTS

E3 tests whether the replay model that agreed with E2 can screen additional checkpoints before deployment. From the existing training archive, we replay 90 previously undeployed candidates under C and C30 on both locked windows (360 CPU scenarios), without retraining. C accepts 67 candidates and rejects 23; C30 accepts 89. Acceptance requires P1+ on both windows. These are replay judgments, not measured success rates.

Selection and execution. The selection rule uses C alone. Accepted candidates are ordered by mean window cost; rejected candidates by total offline and then online deadline misses, with checkpoint identity breaking ties. Within each group, identical complete target trajectories are represented once, and the first three representatives are selected. The

Table 20. E3 checkpoint identities. Names encode training model, seed, and checkpoint index. C rejects N4–N6 because it predicts respectively 3, 6, and 7 offline misses in steady; all six pass recovery in replay.

Arm	Archived checkpoint	C decision
N1	C30_s105_e0065	accept
N2	C30_s105_e0035	accept
N3	C30_s102_e0005	accept
N4	C30_s102_e0080	reject
N5	C_s105_e0080	reject
N6	C_s105_e0070	reject

resulting six C trajectories differ from the previously deployed policies. Table 20 maps the new labels to archived checkpoints; the selected set spans two training seeds. C30 accepts all six but does not determine their selection.

Each checkpoint executes twice per window, with one all2

reference per window: 24 policy runs and two references. A checkpoint is physically feasible only if all four executions satisfy P1+. Recovery and steady execute on separate matched dual-A100 hosts with the same model weights, vLLM 0.9.0 software stack, and accounting specifications. The recorded execution amendment changes work allocation, retaining the six checkpoints, inputs, predictions, and within-window run order. Window and host are therefore confounded; differences between the windows cannot be attributed solely to workload. E3 remains separate from the E2 runtime group.

The launch records bind the frozen scientific files and CPU predictions; actions and replay outcomes match the CPU references. The external Git record was created after physical execution began, so this is a fixed-cohort confirmation, not an externally preregistered study. All 26 executions are technically valid, with no retries or replacement candidates.

Results and interpretation. All 26 runs satisfy P1+. Thus C correctly accepts three checkpoints but falsely rejects the other three; C30 accepts all six correctly on this cohort. There are no physically infeasible checkpoints, and C30 has no rejected candidates, so neither failure-detection performance nor C30’s rejection accuracy can be estimated. The selected cohort and repeated runs also do not provide a population screening-accuracy estimate.

Across the 24 policy runs, feasibility agreement is 18/24 for C and 24/24 for C30: C predicts failure only in the steady cells of N4–N6. Cost MAE is 0.3112 for C and 0.0163 for C30 (Table 21); the two all2 references are excluded from this comparison. These are scenario costs, including the declared synthetic-API route, rather than measured provider charges. The outcome reverses the model ordering observed in E2 on a different cohort and execution context; it does not establish that either fixed model is generally preferable.

In this execution context, the 76 completed in-window launch-to-ready intervals have median 34.78 s (range 32.52–93.01 s), closer to C30’s 30 s assumption than C’s historical 69.219 s value. Two additional late launches are terminated at the observation cutoff before readiness and are excluded from this duration summary as right-censored observations; their window occupancy is charged. These observations do not isolate startup time as the cause of the gate errors. E3 shows why a screening decision itself requires physical validation: a model successful on an earlier cohort can reject feasible policies in a later deployment context.

I RETROSPECTIVE LIFECYCLE-TRAJECTORY DIAGNOSIS

This analysis reuses all nine accepted fixed-H physical runs and the twelve unique O/C/C30/E predictions from the

frozen V3 evaluation: three windows, three physical repeats per window, and four model comparisons per run. The resulting 36 pairings are not new experiments or independent prediction samples. We neither refit a model nor replace the original M1 endpoint. The artifact binds the consumed records to their original acceptance hashes and rechecks all 44,100 decisions across the 21 unique traces.

I.1 Definitions and event alignment

Time is relative to each window start, on $[0, 2100)$ seconds. Physical occupied capacity counts each generation from launch until verified release, clipped to that interval; simulated capacity includes every non-off state, including stopping. Let $n_m(t)$ and $n_r(t)$ be predicted and observed occupied GPU counts. For model m and repeat r , we distinguish

$$A_{m,r} = \left| \int_0^{2100} (n_m(t) - n_r(t)) dt \right|, \quad (2)$$

$$T_{m,r} = \int_0^{2100} |n_m(t) - n_r(t)| dt. \quad (3)$$

Both have units GPU · s. The first is the existing total-occupation absolute error; the second measures trajectory discrepancy without signed temporal cancellation. We integrate the piecewise-constant records exactly, without substituting one-second samples for continuous occupation. Table 23 reports equal-repeat means within each window; Table 24 preserves every pairing.

Targets are compared at all 2,100 one-second decision ticks. The first state-count difference compares the decision-time counts of active, starting, draining/stopping, and off devices; it is not the first difference of all hidden system state. Queue and running observations precede each decision. Routes are compared over all arrival IDs as local, synthetic API, or undispached. For a request whose final routes differ, the recorded first dispatch evidence is a retrospective locator, not a time when both eventual routes were already known. Route differences need not follow target differences.

I.2 An endpoint discrepancy at the cooldown boundary

The frozen C/C30 configurations differ only in the startup endpoint. Their first common starts and subsequent divergences are shown in Table 22. C30 becomes ready after 30 s, but targets stay equal until the 60 s cooldown expires. At that tick C remains starting; its queue-plus-running pressure exceeds the strict up-threshold of eight, and its target rises from one to two. C30 is ready with lower pressure and keeps target one. Both controllers have identical target/cooldown state before this decision; swapping the observed inputs swaps this one-step action. This diagnoses the frozen simulator contrast, not a new physical intervention or a replay of an alternative future trajectory. At the corresponding

Table 21. E3 window costs in scenario USD, with both physical repeats shown. R and S denote recovery and steady. Predictions include any modeled deadline penalty; no physical run incurs one. Each all2 reference satisfies P1+ with cost 4.2000. Costs cover the 2100 s observation window; setup and subsequent cleanup are recorded separately.

Arm	Window	C prediction	C30 prediction	Physical 1	Physical 2	P1+
N1	R	1.9351	1.9351	1.9269	1.9270	2/2
N1	S	1.9767	1.9668	1.9575	1.9579	2/2
N2	R	1.9102	1.8709	1.8698	1.8445	2/2
N2	S	2.0171	1.9517	1.9236	1.9275	2/2
N3	R	1.9720	1.9468	1.9315	1.9322	2/2
N3	S	2.0323	1.9760	1.9617	1.9618	2/2
N4	R	1.9650	1.9621	1.9523	1.9524	2/2
N4	S	2.0689	2.0180	2.0171	2.0172	2/2
N5	R	2.6433	3.6908	3.6664	3.6667	2/2
N5	S	3.0121	3.6548	3.6242	3.6243	2/2
N6	R	2.8257	3.7802	3.7660	3.7580	2/2
N6	S	3.0603	3.9023	3.8758	3.8769	2/2

tick, each of the nine physical repeats has one active device, no starting device, and target one; that is an observational anchor. H does not read the predictor’s startup parameter, and the guard’s separately frozen safety-delay parameters are not intervened on here.

I.3 Counterexamples and temporal cancellation

The six burst/recovery physical repeats first differ from all four models in target at $t = 3$ s, before any window restart. At that tick the models observe no local running request while the physical controllers still do; completion observation has crossed a decision tick. These differences cannot be attributed to window-restart timing. Conversely, steady repeats 1 and 3 have exactly the same 2,100 targets as C30 despite 154 and 152 state-count difference ticks, 48 and 45 route differences, and nonzero occupied-resource errors. Readiness error therefore does not necessarily change a target.

The steady E total MAE is 66.34 GPU · s, close to C30’s 67.18, but its mean trajectory error is 1,390.50 versus 114.32 GPU · s. The same E trace disagrees on 66.73% of target ticks on average. Accurate total occupation can thus hide substantial temporal mismatch; this diagnostic does not retroactively select a different model or alter M1. All C30 pairings retain nonzero occupation error. The largest C30 target residual is burst repeat 2 (637/2100 ticks); its longest mismatch spans [1020, 1144) s. This example is selected by an explicitly post-hoc maximum residual rule after the full audit, alongside all no-target-change cases. The artifact exports every window’s first common-start segment for all models and repeats, as well as the complete startup and unmatched-pair counts.

These observations support checking lifecycle states, actions, and resource trajectories alongside matched-service

fit. Three controlled windows and nine repeats do not establish a universal startup/cooldown law, randomized physical causality, or a general guard guarantee. Software-observed readiness is not a GPU-kernel timestamp, and the synthetic API retains its original quality and billing limitations.

J EXTERNAL-VALIDITY EVIDENCE

The external-validity study (V4 in the artifacts) tests alternative target rules (W1), a second model family (W2), and a second arrival source (W4) under the same deadline and cost definitions. Each test is evaluated separately. The tables distinguish technical validity from strict qualification: the latter keeps the original request denominator, requires at least 99% online on-time completion, every offline job timely, and all requests completed without unfinished or censored work. A late offline completion is not an unfinished request, but neither qualifies. Table 7 accounts for all 93 original V4 identities, including gates and six historical reuses. E1 adds 14 separately accounted formal identities (Section E); requests are not independent experimental repeats.

J.1 Alternative unguarded target rules

The alternative-controller test (W1) asks whether unguarded rules can also complete all work on time. B-HPA is an HPA-inspired queue-target controller with a stabilization interval, not an implementation of Kubernetes HPA. B-PRED applies a causal Holt queue predictor; B-MMC selects a capacity from the past arrival-rate estimate and the frozen E service lookup. The three policies share the two-instance limit, lifecycle semantics, one-second observation tick, and 60-second action cooldown. None uses the V3 deadline guard, observes future arrivals, or trains a learned scheduler. No registered development candidate qualified, so all three use

Table 22. First C/C30 startup contrast in all three windows. Times are seconds from window start; pressure is queued plus locally running requests. The first target difference occurs at common start plus 60 s in each window.

Window	Common start	C30 ready	C ready	First target difference	Pressure C/C30
Burst	195	225	264.219	255	13/1
Recovery	126	156	195.219	186	14/2
Steady	171	201	240.219	231	21/4

Table 23. All window/model means over three physical repeats. Total MAE is mean $A_{m,r}$; trajectory error is mean $T_{m,r}$, both in GPU · s. Target disagreement uses 2,100 ticks per pair; route disagreement uses all arrival IDs in that window, including undispached requests.

Window	Model	Total MAE	Trajectory error	Target diff. (%)	Route diff. (%)
Burst	O	234.48	457.24	15.17	27.95
Burst	C	847.60	1369.86	49.63	27.61
Burst	C30	83.08	268.96	10.33	15.26
Burst	E	70.55	237.76	10.33	15.71
Recovery	O	244.08	421.31	11.71	27.75
Recovery	C	864.20	1302.19	43.92	35.24
Recovery	C30	78.30	138.31	3.24	2.20
Recovery	E	66.95	99.02	3.24	4.19
Steady	O	1151.97	1517.83	59.40	42.84
Steady	C	964.14	1621.50	73.68	38.87
Steady	C30	67.18	114.32	1.03	4.99
Steady	E	66.34	1390.50	66.73	49.45

their predeclared center fallback; they are not presented as optimized best-in-family implementations. Section C.1 records their grids, locks, and semantic validation.

All three baselines complete all nine formal runs with timely online and offline work (Table 25). Their equal-window scenario costs are 2.2305, 4.0085, and 2.7861, respectively. Each exceeds the historical V3 all1 reference, showing that timely completion alone does not identify the lowest-cost policy. All three succeed without the guard; adding the guard to these controllers remains untested. B-PRED’s per-run online p99 is 1.58–2.70 seconds across these windows, whereas B-HPA and B-MMC reach 15–19 seconds in steady/recovery. Thus the higher-cost predictor also exposes a latency–resource trade-off; Table 8 reports each window’s three-repeat range.

This comparison has a runtime boundary. W1’s historical cells 001/002 and its r14 continuation belong to different runtime groups, and the new arms are not randomized same-runtime pairs with the frozen V3 references. This deployment confound prevents a causal comparison with H+guard.

J.2 A same-runtime supplementary comparison

E1 addresses the runtime confound with fresh H+guard and B-HPA runs on the known steady and recovery inputs. Both policies meet every deadline; H+guard reduces window cost by 32.60% and 49.36%, respectively, while routing

more online work to the synthetic API (Table 26). This is a comparison of complete policies, including their dispatch rules. Section E provides the shared-runtime contract, all 14 run identities, phase costs, and resource–cost derivation.

The benefit depends on available latency margin and external-service prices: lower occupancy comes with more lifecycle events and queue exposure, and the API break-even multipliers are 3.08 and 11.99 at the registered GPU price. These known-input results bound the guard’s resource trade-off rather than establishing performance on new workloads.

J.3 A second model family

The cross-model test (W2) uses Llama-3.1-8B-Instruct in bfloat16 with two independent TP1/PP1/DP1 instances. The 24-cell matrix contains six historical all2 gate cells reused once and 18 fresh cells, with two repeats per arm and window. All 24 cells are technically valid; the original scientific acceptance count is 23/24, whereas the stricter request-completeness count is 21/24 (Table 27). These denominators answer different questions and are not interchangeable.

H leaves four offline jobs unfinished in each recovery repeat (cells 042/043); the denominators remain 120, not the 116 observed completions. The guard completes all jobs but one recovery repeat (cell 049/r2) has only 119/120 timely offline completions. The registered W2-P guard-feasibility claim therefore fails. H’s recovery failure recurs across families,

Table 24. All 36 model/physical-repeat pairings. Errors are GPU · s; target differences are counts out of 2,100 ticks. A dash means targets never differ, not a missing record. Route counts retain the full arrival denominator. The twelve unique predictions are reused across the nine physical runs.

Window	Model	Rep.	Total error A	Trajectory T	Target diff.	First diff. (s)	Route diff./arrivals
Burst	O	1	300.54	304.39	147	3	65/297
Burst	O	2	103.59	764.07	662	3	117/297
Burst	O	3	299.30	303.26	147	3	67/297
Burst	C	1	913.66	1347.35	1056	3	76/297
Burst	C	2	716.71	1416.14	1015	3	90/297
Burst	C	3	912.42	1346.08	1056	3	80/297
Burst	C30	1	53.54	53.54	7	3	2/297
Burst	C30	2	143.41	701.03	637	3	128/297
Burst	C30	3	52.30	52.30	7	3	6/297
Burst	E	1	4.48	6.53	7	3	6/297
Burst	E	2	201.43	698.53	637	3	122/297
Burst	E	3	5.72	8.24	7	3	12/297
Recovery	O	1	305.21	335.21	187	3	122/454
Recovery	O	2	122.16	593.86	364	3	133/454
Recovery	O	3	304.85	334.85	187	3	123/454
Recovery	C	1	925.33	1233.56	895	3	156/454
Recovery	C	2	742.28	1440.00	977	3	167/454
Recovery	C	3	924.97	1233.01	895	3	157/454
Recovery	C30	1	52.21	52.21	9	3	10/454
Recovery	C30	2	130.84	310.86	186	3	11/454
Recovery	C30	3	51.85	51.85	9	3	9/454
Recovery	E	1	5.81	8.75	9	3	21/454
Recovery	E	2	188.86	279.36	186	3	16/454
Recovery	E	3	6.17	8.95	9	3	20/454
Steady	O	1	1193.59	1538.83	1255	99	686/1564
Steady	O	2	1068.73	1475.71	1232	99	635/1564
Steady	O	3	1193.58	1538.95	1255	99	689/1564
Steady	C	1	1005.76	1623.09	1548	231	596/1564
Steady	C	2	880.91	1618.33	1546	231	635/1564
Steady	C	3	1005.76	1623.10	1548	231	593/1564
Steady	C30	1	76.70	76.70	0	—	48/1564
Steady	C30	2	48.15	189.57	65	969	141/1564
Steady	C30	3	76.70	76.70	0	—	45/1564
Steady	E	1	74.18	1388.10	1400	232	784/1564
Steady	E	2	50.68	1395.18	1404	232	755/1564
Steady	E	3	74.17	1388.23	1400	232	781/1564

Table 25. Alternative controllers (W1) under aligned deadline and cost definitions. Occupied time is in GPU seconds; cost is window scenario USD. Saving is against the historical V3 all1 equal-window cost. The arms and historical reference use different runtimes.

Policy	strict	occupied	cost	saving (%)
B-HPA	9/9	2199.5	2.2305	-6.06
B-PRED	9/9	4007.5	4.0085	-90.62
B-MMC	9/9	2752.6	2.7861	-32.49

while the guard loses its perfect on-time record. With two repeats per arm/window, this test identifies a transfer failure but cannot establish reliability across model families.

Table 26. E1: means of three fresh repeats per arm/window; all pass P1+. Cost is window scenario USD, and API share is percent of all online arrivals.

Window	Policy	Cost	API (%)
steady	B-HPA	2.4685	6.44
steady	H+guard	1.6638	51.64
recovery	B-HPA	2.0714	11.68
recovery	H+guard	1.0489	59.38

The within-Llama model comparison is also negative. On fixed H, O_x and E_x both have equal-window $L = 0.00421052$ and occupied MAE 25.1458 GPU seconds: neither improves over the other. The registered gate requires at least 20% L reduction *and* each window’s occupied MAE no

Table 27. W2 Llama results. Completed and timely columns aggregate offline jobs only to expose missing or late work; 720 = six runs $\times$ 120 jobs. Runs, not the 720 jobs, determine the strict qualification count.

Policy	strict	completed	timely	boundary
all2	6/6	720/720	720/720	none
all1	6/6	720/720	720/720	none
H	4/6	712/720	712/720	8 unfinished
H+guard	5/6	720/720	719/720	1 late

worse than $O_x + 21$; passing the latter does not rescue zero improvement. The inherited 20% occupied-improvement diagnostic also fails. These locks use development median/max aggregate service lookups with identical lifecycle endpoints, not a new Llama demonstration of the Qwen endpoint-adaptation gain. M2 passes on completed-local common support (minimum 206 requests, maximum MAE 0.02964 s), but H and guard cover only 51.24% and 58.90% of all arrivals on an equal-window basis. This service-subset result does not establish whole-stream or controller fidelity; the supplement reports coverage and target disagreement separately.

J.4 Same-environment lifecycle refit

W3 asks whether a bounded same-environment refit resolves the original V–E comparison’s environment caveat. It selects V' by loss on two preformal development canaries; a separate preformal drift probe constrains lifecycle lower bounds. It freezes the model before CPU replay and evaluates all 36 existing physical records without new GPU runs. On fixed H, equal-window L falls from 0.01352738 to 0.01284857, a 5.02% reduction below the registered 10% threshold, while every window satisfies the occupied-MAE safety line $V' \leq E + 21$. Steady-window L worsens despite the aggregate reduction. The result is a scientific negative, not evidence that refitting has no effect. Only lifecycle scalars are eligible for fitting; service, setup, and guard safety remain fixed. The researchers already knew V3’s aggregate results, although the fitting code never consumed formal outcomes. This is a bounded retrospective replay, not a newly blinded experiment, and it does not erase the historical V–E caveat.

J.5 Second arrival source and a load boundary

The Azure conversation source associated with Splitwise (Patel et al., 2024) contains 19,366 rows over about 58.4 minutes. W4 uses timestamps from its first and last non-overlapping 25-minute windows. Replay uses fixed-corpus prompts and controlled token budgets, with the V3 synthetic API contract. This does not replay the original prompt

content or generated responses. The original full-strength all2 gate is technically valid but scientifically negative in all six runs (Table 9); the other 18 planned policy cells are NOT_RUN_GATE_STOP, not missing successes. Equal-window mean online timely completion is only 66.2% (39.6–93.1% across runs), and offline timely completion is 29.9%. No feasible-saving claim is made at that intensity.

The single sequential W4-L amendment fixes integer SHA thinning at online intensities 0.75, 0.50, then optionally 0.25. It preserves retained row identities, arrival times, deadlines, and all 120 offline jobs per window. At 0.75, all work eventually completes, but only 113/120 front-window and 79/120 back-window offline jobs are timely in each repeat. At 0.50, all six all2 gate runs pass: this is the first feasible tested level. The subsequent 24 fresh runs use that locked level, four policies, two windows, and three repeats. The six successful gate runs are not merged into the fresh all2 denominator. The optional 0.25 level was not added.

The fresh comparison separates cheap failure from feasible saving (Figure 4 and Table 10). all1 qualifies in 0/6 runs and H in 3/6: H’s front repeats 1 and 3 each censor two offline jobs, while back repeat 2 censors four. These eight unfinished jobs fail the strict completion criterion despite the recorded `contract_feasible` flag. The guard qualifies in 6/6 and reduces equal-window occupied time from 4200.000 to 2992.407 GPU seconds (28.75%), window cost from 4.2000 to 3.0675 (26.96%), and full-deployment cost from 4.3204 to 3.1366 (27.40%). These percentages are ratios of equal-window means, not averages of paired per-run percentages. The official aggregate `NEGATIVE_RESULT`, driven by all1, is preserved. The time decomposition further limits the economic interpretation: guard cost falls only 6.81% during the first 1,500 seconds, and about 81.96% of the absolute whole-window saving arises in the final 600 seconds. The 26.96% headline must therefore not be extrapolated to sustained high-load serving efficiency. Figure 4 makes the concentration explicit; the supplement retains the numerical segment ledger.

Thinning reuses observed source windows; intensity was selected by the registered all2 feasibility gate, not outcome-independent sampling. The result is therefore load-conditioned, not a new held-out trace. Offline arrivals continue during [1500, 1800), and the back window retains an online arrival at the closed 1500-second endpoint, so source-window timing and ledger segments are reported separately. These results do not estimate Azure’s production QoS, required fleet size, model quality, energy use, or bills.

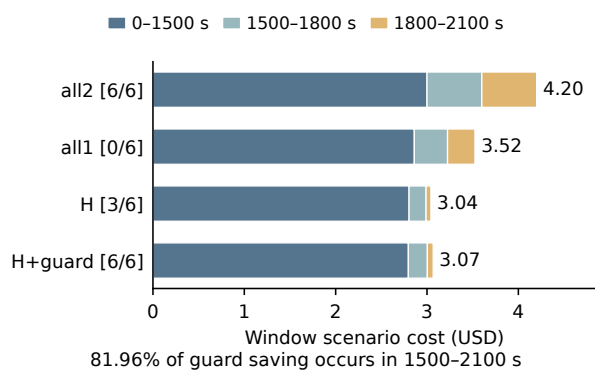


Figure 4. W4-L costs by window-time segment. Bars average three repeats per window, then weight the two windows equally; brackets give strict passes out of six. Lower cost in failed runs is not feasible saving. Of the guard’s saving against all2, 81.96% occurs in the final 600 seconds.

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